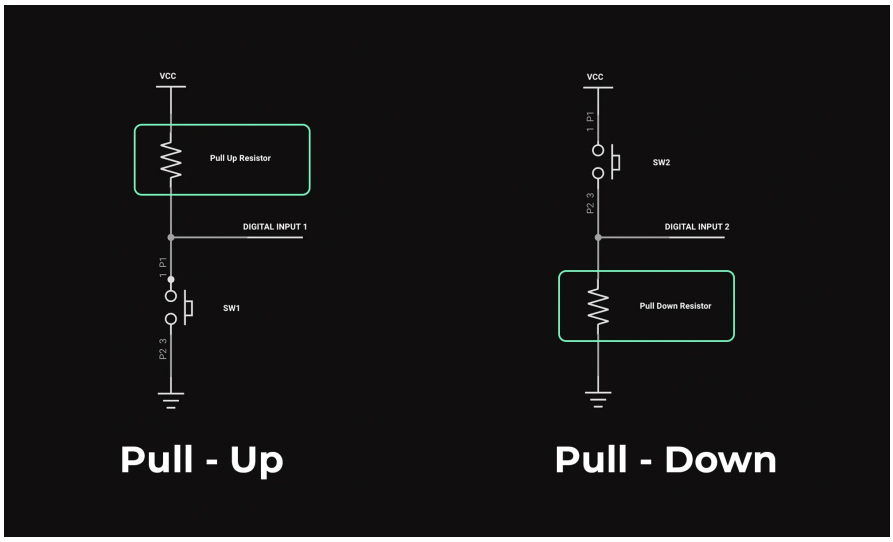
PW 2

1 PullUp & PullDown Resistance

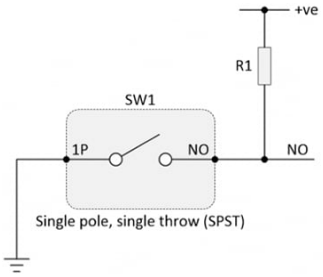
- **PullUp Resistance** : the resistance is linked to the **VCC** and the switch to **GND**, the active state is `LOW`
- **PullDown Resistance** : the resistance is linked to the **GND** and the switch to **VCC**, the active state is `HIGH`

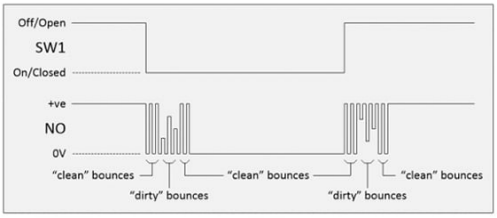


Note that all the pins in the STM32F411 have resistance both ways (PullDown & PullUp)

2 Debouncing

When pressing mechanical switch there is dirty noise fluctuation between the hold and release to avoid those we add small delay to have clean `LOW`, `HIGH`





3 Code To Turn On/Off LED PINC13 Using Button PinA0

We will use both methods PullUp and PullDown :

3.1 PullUp

pullup.ino

```
#define LED_PIN PC13            // declare led Pin C13
#define BUTTON_PIN PA0          // declare button Pin A0
bool lastButtonState = HIGH;    // initial Button Off is High (PullUp)
bool ledState = HIGH;           // initial LED Off is High (sinking current)

void setup(){
  pinMode(LED_PIN, OUTPUT);                   // set Led pin to output
  pinMode(BUTTON_PIN, INPUT_PULLUP);          // set Button pin to input pullup
  digitalWrite(LED_PIN, ledState);            // set initial state of Led pin
  Serial.begin(115200);                       // serial monitor will send 115200 bits per second
  while(!Serial) {                            // loop as long as serial not ready
    delay(500);                               // delay 500 ms
    Serial.println("STM32F411 started");      // print init message
    Serial.println("Press the button to toggle the LED");
    }
}
void loop() {
  bool currentButtonState = digitalRead(BUTTON_PIN);          // read button state
  if(lastButtonState == HIGH && currentButtonState == LOW){   // if button was clicked (off then on)
    ledState = !ledState;   // invert state of led
    digitalWrite(LED_PIN,ledState); // write on led pin

    if(ledState == LOW){                // if led LOW (means ON)
      Serial.println("LED ON");         // print led on
    } else {
      Serial.println("LED OFF");        // print led off
    }

  }
  lastButtonState = currentButtonState;   // update last state
  delay(200);                             // delay for 200ms for debouncing
}
```

3.2 PullDown

pulldown.ino

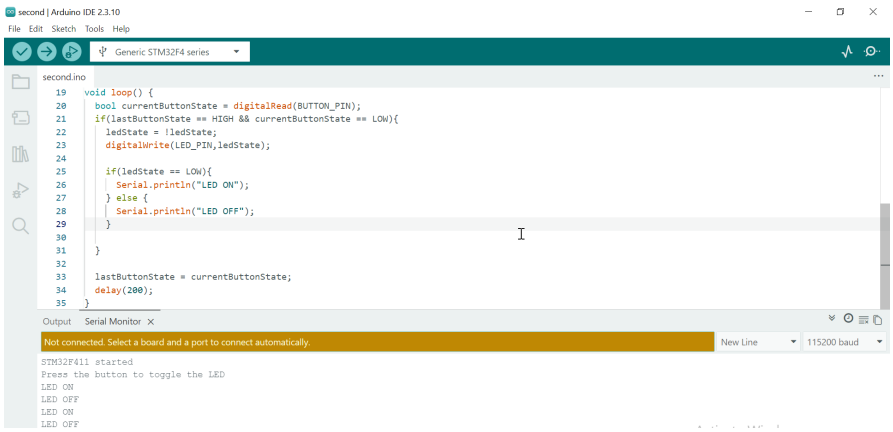
```
#define LED_PIN PC13            // declare led Pin C13
#define BUTTON_PIN PA0          // declare button Pin A0
bool lastButtonState = LOW;     // initial Button Off is Low (PullDown)
bool ledState = HIGH;           // initial LED Off is High (sinking current)

void setup(){
  pinMode(LED_PIN, OUTPUT);                   // set Led pin to output
  pinMode(BUTTON_PIN, INPUT_PULLDOWN);        // set Button pin to input pulldown
  digitalWrite(LED_PIN, ledState);            // set initial state of Led pin
  Serial.begin(115200);                       // serial monitor will send 115200 bits per second
  while(!Serial) {                            // loop as long as serial not ready
    delay(500);                               // delay 500 ms
    Serial.println("STM32F411 started");      // print init message
    Serial.println("Press the button to toggle the LED");
    }
}
void loop() {
  bool currentButtonState = digitalRead(BUTTON_PIN);          // read button state
  if(lastButtonState == LOW && currentButtonState == HIGH){   // if button was clicked (off then on)
    ledState = !ledState;   // invert state of led
    digitalWrite(LED_PIN,ledState); // write on led pin

    if(ledState == LOW){                // if led LOW (means ON)
      Serial.println("LED ON");         // print led on
    } else {
      Serial.println("LED OFF");        // print led off
    }

  }
  lastButtonState = currentButtonState;   // update last state
  delay(200);                             // delay for 200ms for debouncing
}
```

4 Serial Monitor Output



5 Execution

- **PullUp** : The integrated button work normally because it is linked to the **GND** therefore we are respecting the PullUp
- **PullDown** : The integrated button isn't working due to the fact button is linked to the **GND** not the **VCC** therefore not respecting the PullDown

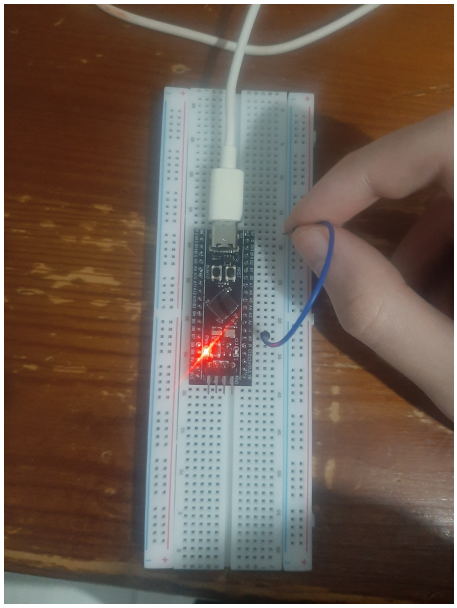
6 Breadboard

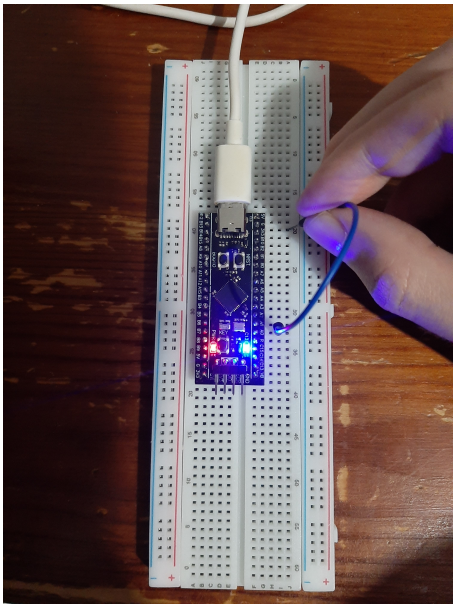
since i don't have an external button i'll be using male to male jumper to simulate a button, as always put the STM32F411 in the middle then :

- **PullUp** : One side linked to pin A0 (button pin) and other side to **GND**
- **PullDown** : One side linked to pin A0 (button pin) and other side to **3.3V**

Each time other side is plugged as if pressing button

6.1 PullUp





6.2 PullDown

